



CRYPTOCHAIN ANALYZER DASHBOARD

Hash Functions and Blockchain - Final Report

1. Project Overview

This project implements a real-time Python dashboard that connects to public Bitcoin APIs and analyses live cryptographic metrics from the Bitcoin network. The dashboard is organised around the four required modules: Proof of Work monitoring, block header analysis, difficulty history, and an AI anomaly detection component.

The objective is not only to display blockchain data, but also to connect it with the cryptographic concepts studied in class: SHA-256, double hashing, Proof of Work, compact target encoding, difficulty, nonce search, Merkle roots, and transaction inclusion through the mempool.

2. Cryptographic Metrics Displayed

M1 - Proof of Work Monitor: The dashboard displays the latest Bitcoin block height, hash, nonce, number of transactions, current difficulty and a theoretical hash rate estimate. Difficulty represents how hard it is to find a block hash below the current mining target. The hash rate is approximated using: $\text{difficulty} \times 2^{32} / 600$. Therefore, it should be understood as an estimate based on the expected 600-second block interval, not as a direct measurement of mining hardware.

The dashboard also plots the time between recent blocks and compares it with the Bitcoin target of 600 seconds. Individual block intervals are not expected to be exactly ten minutes because mining is probabilistic: miners repeatedly test nonces and other block header variations until the resulting hash is below the target. Therefore, block discovery behaves like a random process even though the long-term protocol target is one block every ten minutes.

M2 - Block Header Analyzer: The dashboard retrieves the latest block header fields: version, previous block hash, Merkle root, timestamp, bits and nonce. It serialises the 80-byte block header in Bitcoin little-endian format and recomputes: **SHA256(SHA256(header))** locally using Python hashlib. The locally computed hash is then compared with the API hash. The Proof of Work check is valid when the numerical value of the block hash is lower than or equal to the target decoded from bits.

The bits field is especially important because it does not store the difficulty directly. It stores the mining target in compact form: an exponent and a coefficient. The dashboard decodes bits into the full 256-bit target, calculates the corresponding difficulty, and counts the leading zero hexadecimal characters and leading zero bits in the displayed block hash. This connects the visual appearance of valid block hashes with the underlying Proof of Work threshold.

M3 - Difficulty History: The dashboard shows recent historical difficulty values from Blockchain.com chart data and highlights significant visible variations in the sampled series. Bitcoin adjusts mining difficulty periodically to keep the average block interval close to 600 seconds. In the dashboard, the visualisation is used to interpret the trend in mining difficulty and to show how changes in network mining power are reflected in the target threshold.

3. Data Sources and Dashboard Functionality

Source	Use in the dashboard
Blockchain.com	Latest block, raw block information, difficulty chart data.
Blockstream API	Block header fields required for local Proof of Work verification.
Mempool.space	Mempool size, recent pending transactions and recommended fee rates.

The dashboard is implemented with Streamlit and Plotly. It includes automatic refresh using streamlit-autorefresh, configurable recent block windows, configurable anomaly thresholds, and basic API error handling so that external API failures do not crash the interface without explanation.

4. AI Component: Anomaly Detection on Block Times

The selected AI approach is an anomaly detector for abnormal Bitcoin inter-block times. This option was chosen because it is directly connected to Proof of Work theory and to the M1 block-time analysis.

In Bitcoin, block discovery can be approximated as a probabilistic process. A common theoretical baseline is that inter-block times follow an exponential distribution with an expected mean of 600 seconds. The model analyses recent real block intervals and flags very short or very long intervals as potentially anomalous when their tail probability under this exponential baseline is below the selected threshold.

Data used: The model uses recent Bitcoin block timestamps retrieved from the Blockstream API. Consecutive timestamps are converted into positive inter-block intervals in seconds. These intervals are the input observations for the anomaly detector.

Detection criterion: For each interval x , the dashboard computes lower-tail and upper-tail probabilities using an exponential baseline with expected mean 600 seconds. Very small intervals are unusual because a block was found much faster than expected; very large intervals are unusual because the network took much longer than expected. The final anomaly score is the smaller of the two tail probabilities. If that score is below the selected threshold, the interval is labelled anomalous.

Evaluation: There is no reliable labelled ground truth indicating whether each recent Bitcoin block interval is truly anomalous or not. For this reason, the model is evaluated statistically rather than with supervised metrics such as accuracy or F1-score. The dashboard reports: total anomalies, anomaly rate, fast anomalies, slow anomalies, the expected mean of 600 seconds, the selected threshold, and the observed mean interval in the current sample. This makes the model transparent and allows the user to inspect how sensitive the detector is to the threshold.

Limitations: A detected anomaly does not automatically mean that there is an attack or manipulation. Bitcoin block discovery is random, so extreme intervals can occur naturally. The model also analyses only a recent sample of blocks, which means the result changes over time and depends on the selected window size. The detector should therefore be interpreted as an exploratory statistical warning system, not as a definitive security classifier.

5. Additional Research Element: Mempool and Fee Market

To go beyond the minimum required modules, the dashboard includes an extra mempool overview. The mempool is the set of valid transactions waiting to be included in future blocks. The extra section shows the number of pending transactions, total virtual size, recommended fee rates and a distribution of recent mempool transaction fee rates in sat/vByte. This connects block production with transaction demand: when the mempool becomes more congested, users generally need to attach higher fees to obtain faster confirmation priority.

6. Main Results

The final dashboard successfully loads live Bitcoin data, verifies the latest block Proof of Work locally, displays block timing behaviour, visualises historical difficulty data, runs a statistical anomaly detector on real block intervals, and adds a mempool fee-market view. The strongest cryptographic part is the local reconstruction of the 80-byte block header and the double SHA-256 verification, because this demonstrates that the dashboard is not merely displaying API values but actually validating the Proof of Work mechanism. The project also shows an important practical lesson: blockchain metrics must be interpreted carefully. A high difficulty means a lower target and more expected work; leading zeros are only a visual consequence of the numerical threshold; the nonce is not the hash itself; and abnormal block intervals are not necessarily malicious because Proof of Work mining is inherently probabilistic.

7. References

Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. <https://bitcoin.org/bitcoin.pdf>

Blockstream Esplora API Documentation. <https://github.com/blockstream/esplora/blob/master/API.md>

Blockchain.com Charts API and Network Difficulty documentation. https://www.blockchain.com/api/charts_api
<https://www.blockchain.com/charts/difficulty>

Mempool.space API documentation and public mempool data. <https://mempool.space/api>

Bowden, R., Keeler, H. P., Krzesinski, A. E., & Taylor, P. G. (2018). *Block arrivals in the Bitcoin blockchain*. arXiv:1801.07447.

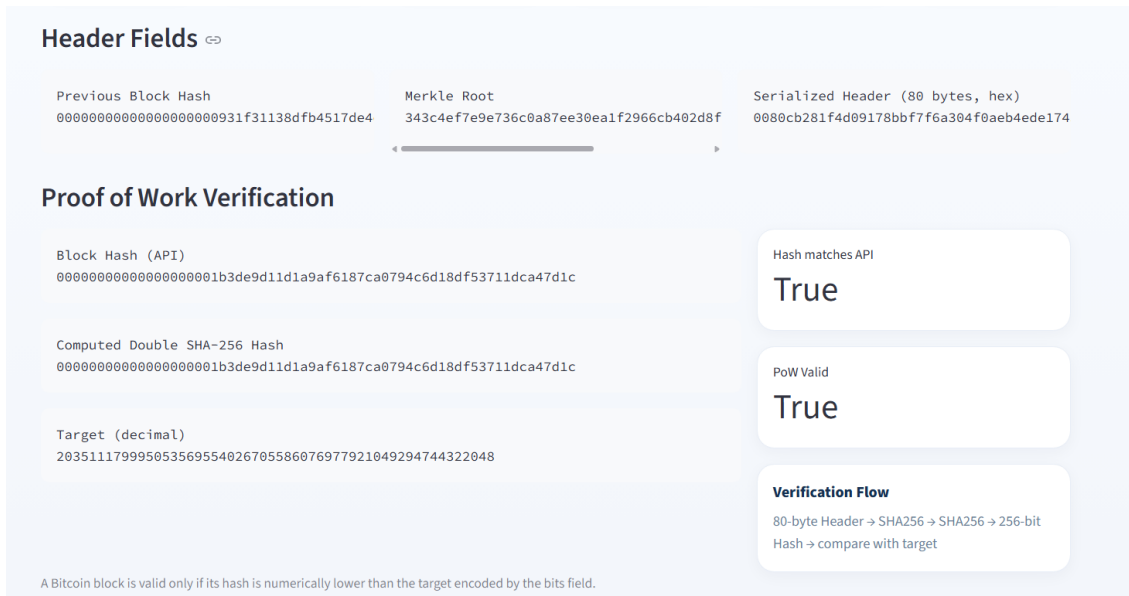


Figure 1. Local Proof of Work verification for the latest Bitcoin block using double SHA-256 and target comparison.